System Design

Component implementation

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How to approach a new codebase

1. Understand $\rightarrow$ 2. Assimilate $\rightarrow$ 3. Implement $\rightarrow$ 4. Verify

1. Understand

- **Start with the README.** Set up your development environment. Check the main folder structure.
- **Identify the major components** (database, API, front end) and the main technologies used.
- **Connect the dots.** Choose one simple feature and trace its path through the code. Follow the data: where does it start, where does it go, what happens to it?
- **Read** comments, internal documentation, and file names to see how the pieces fit together.

Goal: build a mental model of the code's architecture and logic.

2. Assimilate

- **Become a "code chameleon."** Your first job is to blend in. Mimic the style you see: naming conventions, comment format, project structure.
- **Trust your intuition.** If the language is new, don't panic. Use your general programming knowledge to follow the logic; the syntax comes as you work.

Goal: learn the code's patterns and local dialect before you write anything new.

3. Implement

- **Make a small, controlled change.** It is the fastest way to learn.
- Start with a tiny, well-defined bug fix or a very simple feature.
- This forces you to engage with the code, its patterns, and its dependencies.

Goal: move from passive observer to active contributor.

4. Verify

Prove your change works, and that you didn't break anything else.

- Run the project's existing test suite: unit tests, integration tests.
- Try the change by hand where you can. In HW3 there is no app to click through; the tests and the simulation are how you run the service.

Goal: confirm your implementation is correct and the codebase is still stable.

Homework 3: implement the Machine Service

Complete the `ApiHandler` class in `src/handler/api.ts` of the starter repository.

Submit:

- all source code, and
- a report (PDF) with a screenshot of the service's performance statistics and a short analysis.

Full handout: `assignments/hw3-machine-service/HOMEWORK.md`